5. Write a short note on vector product between two vectors.
6. How do you deduce that two vectors are perpendicular?
7. Define displacement and distance.
8. Define velocity and speed.
9. Define acceleration.
10. What is the difference between velocity and average velocity.
11. Define a radian?
12. Define angular displacement and angular velocity.
13. What is non uniform circular motion?
14. Write down the kinematic equations for angular motion.
15. Write down the expression for angle made by resultant acceleration and radius vector in the non uniform circular motion.

III. Long Answer Questions

1. Explain in detail the triangle law of addition.
2. Discuss the properties of scalar and vector products.
3. Derive the kinematic equations of motion for constant acceleration.
4. Derive the equations of motion for a particle (a) falling vertically (b) projected vertically
5. Derive the equation of motion, range and maximum height reached by the particle thrown at an oblique angle θ with respect to the horizontal direction.
6. Derive the expression for centripetal acceleration.
7. Derive the expression for total acceleration in the non uniform circular motion.

IV. Exercises

1. The position vectors particle has length 1m and makes 30° with the *x*-axis. What are the lengths of the x and y components of the position vector?

   [Ans: $l_x = \frac{\sqrt{3}}{2}, l_y = 0.5$]
2. A particle has its position moved from $\vec{r}_1 = 3\hat{i} + 4\hat{j}$ to $\vec{r}_2 = \hat{i} + 2\hat{j}$. Calculate the displacement vector $(\Delta\vec{r})$ and draw the $\vec{r}_1$, $\vec{r}_2$ and $\Delta\vec{r}$ vector in a two dimensional Cartesian coordinate system.

   [Ans: $\Delta\vec{r} = -2\hat{i} - 2\hat{j}$]
3. Calculate the average velocity of the particle whose position vector changes from $\vec{r}_1 = 5\hat{i} + 6\hat{j}$ to $\vec{r}_2 = 2\hat{i} + 3\hat{j}$ in a time 5 second.

   [Ans: $\vec{v}_{avg} = -\frac{3}{5}(\hat{i} + \hat{j})$]
4. Convert the vector $\vec{r} = 3\hat{i} + 2\hat{j}$ into a unit vector.

   [Ans: $\hat{r} = \frac{(3\hat{i} + 2\hat{j})}{\sqrt{13}}$]
5. What are the resultants of the vector product of two given vectors given by

   $\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{B} = 5\hat{i} + 3\hat{j} - 4\hat{k}$?

   [Ans: $5\hat{i} + 21\hat{j} + 22\hat{k}$
6. An object at an angle such that the horizontal range is 4 times of the

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